

Tangent substitution - 3.3

Example: Find $\int \frac{1}{(9+x^2)^{3/2}} dx$. $x = 3 \tan \theta$
 $dx = 3 \sec^2 \theta d\theta$

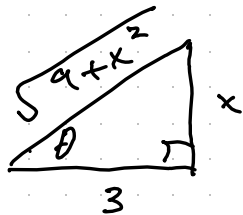
$$= \int \frac{1}{(9+9\tan^2\theta)^{3/2}} \cdot 3 \sec^2 \theta d\theta$$

$$= \int \frac{1}{(9\sec^2\theta)^{3/2}} \cdot 3 \sec^2 \theta d\theta$$

$$= \int \frac{1}{(3\sec\theta)^3} \cdot 3 \sec^2 \theta d\theta = \int \frac{1}{27\sec^3\theta} \cdot 3 \sec^2 \theta d\theta$$

$$= \int \frac{1}{9\sec\theta} d\theta = \frac{1}{9} \int \cos\theta d\theta = \frac{1}{9} \sin\theta + C$$

$$\tan \theta = \frac{x}{3}$$



$$\frac{1}{9} \sin \theta + C$$

$$= \frac{x}{9\sqrt{9+x^2}} + C.$$